Dipartimento di Ingegneria e Scienza dell'Informazione

- KnowDive Group -

Schools and Kindergartens of Trento

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1. Introduction

Reusability is one of the main principles in the Knowledge Graph (KG) development process defined by iTelos. The KG project documentation plays an important to enhance the reusability of the resources handled and produced during the process. A clear description of the resources, the process (and sub processes) developed and evaluation at each step of the process provides a clear understanding of the project, thus serving such an information to external readers for the future exploitations of the project's outcomes. The current document aims to provide a detailed report of the project developed following the iTelos methodology. The report is structured, to describe:
- Section 2: Definition of the project's purpose and related information gathering.
- Sections 3, 4, 5, 6: The description of the iTelos process phases and their activities, divided

by knowledge and data layer activities, as well as the evaluation of the resources produced in terms of fit for the chosen purpose.
- Section 7: Conclusion and open issues summary.

This document provides a comprehensive overview of the development process for the KGE project. It outlines the key stages, technical decisions, and implementation details that shaped the project from initial concept to completion. The code and the document itself can be found on GitHub at the following link

2. Purpose Definition

2.1. Informal Purpose

The project originated from a simple but rather useful informal idea: "to create a service that helps users find schools in the Trentino region by municipality and geographical position and explore details such as school type, number of classes and students and graduation rates."

From this objective, the project's **Domain of Interest** can be defined as all information related to educational institutions in Trentino and the municipalities of the Autonomous Province of Trento. This includes data about each institution's facilities, the number of students and classes, and the graduation rate. The facilities included in this study are kindergartens, elementary and secondary schools (universities have been left out). The information related to these facilities as well as all the municipalities belonging to the autonomous region of Trento is part of the project domain. Moreover additional information such as the number of students and classes and graduation rates from each school is included.

Ultimately, the goal is to build a knowledge graph that serves as a unified repository of educational information for Trentino. This system should make it easier for both residents and visitors to discover relevant insights about the region's educational opportunities. The following sections further detail the project's purpose through Personas and Scenarios.

2.2. Scenarios Definition

In this section we describe possible **scenarios**, meaning carefully constructed fictional events that represent how users might realistically interact with and benefit from our knowledge graph. By following these imagined use cases, readers can better understand the kinds of questions the knowledge graph can answer.

2.2.1. Schools Nearby

It's common for parents to look for kindergartens either near their home municipality or the one of their workplace. Municipalities that are close enough to be reached by car in an acceptable amount of time could also be considered. Usually this research involves also looking for the school website and note down some contact email or telephone.

2.2.2. School research

For pupils choosing the best high school for them is a very important decision to make. Distance is for sure an important factor, but other parameters can come into play, such as being public or private, number of students and graduation rates. Consulting the educational offer on the school website is also fundamental. Some institutes may also offer the full educational career from primary to high school all in one place and this may be convenient for some parents.

2.2.3. Census

A public officer wants to map every educational facilities across the Trentino region in order to understand their distribution, identify underserved areas, and support evidence based planning for future investments in education.

2.2.4. Quality assessment

A civil servant is tasked to conduct a yearly quality assessment. By monitoring available statistics such as the number of students and classes and admission rates, the experts should be able to check if all the schools meet the desired standards.

2.3. Personas

At the start of this section, we highlighted how essential persona analysis is for grasping the project's target domain and clarifying its goals. Personas serve as abstractions for real users: they capture their motivations, goals, and defining characteristics. When they accurately reflect real world behaviors, personas function as a reliable reference point for guiding both design and development choices. In this work, the personas represent the types of users who will eventually interact with the output of the knowledge graph engineering process. This subsection therefore outlines several user categories, each engaging with the system for their own reasons. Although all users may share the basic task of searching for schools in the Trentino region, their individual objectives can vary, which leads them to use the system, and interpret its results, in different ways.

2.3.1. Annamaria Verdi

Annamaria Verdi is a mother of a child who is going to attend elementary school soon. She currently resides in Rovereto and she doesn't have a car, so she would like to look up all the educational facilities in her vicinity. She also would like to look up the school website to peek at the photos, read the educational offer and ensure her child has everything he needs.

2.3.2. Fabrizio Alighieri

Fabrizio Alighieri is a 14 year old teenager who needs to decide which high school ("Liceo") is the best for him. He wants to focus his search on all the private schools near the city of Trento. He is also interested in contacting the school's secretary office to see if it is possible to visit the school in person during the open days.

2.3.3. Luciano Accorti

Luciano Accorti is a retired high school professor. He would like to launch a small private institute to make education more accessible for his municipality. Luciano needs a way to check all the schools in the vicinity, the type (public / private, primary / secondary) and how many students they have to see if his idea is viable and to carefully plan all the details.

2.3.4. Viviana Rossi

Viviana Rossi is an athlete, she has won the gold medal at the last winter olympics. She is a regular guest at the Sport Festival of Trento and now she wants to speak in high schools around Trentino. Viviana needs a handy way to make a tour, selecting all the schools with enough students.

2.3.5. Giuseppe Veronesi

Giuseppe Veronesi is a public officer. He needs to organize the new math competition for the Trento province. For this task he needs to divide the territory into smaller regions in order to define the selection phases. The regions must be balanced by number of schools and students.

2.4. Competency Questions

Once the personas and their needs have been defined it can be more clearer how the project is intended to be used. We came up with a series of realistic Competency Questions based on the possible needs of the user base. These questions will be used to test the reliability and exhaustiveness of the final knowledge graph implementation.

1. Give me the complete list of schools from the Rovereto municipality.
2. Give me the number of high schools in San Michele All'Adige.
3. Give me the list of kindergartens from Lavis.
4. Give me the list of primary schools from Trento and nearby municipalities.
5. Give me the schools in the municipalities with the highest altitude
6. Give me the schools from the least populated municipalities
7. Give me the list of schools with the highest admission rates to the next year
8. Give me the list of private high schools from Pergine Valsugana, ordered by graduation rates.
9. Give me the schools with the lowest number of students.
10. Give me the schools with the highest number of students.

11. Give me the list of elementary schools within a 20km range from Trento

2.5. Information Gathering

This section describes, at top level, the work behind the information gathering phase. Three main sources of raw data have been selected in order to build the final knowledge graph.

2.5.1. Wikipedia

In order to obtain all the municipalities from the Trentino region, wikipedia was selected as a source. Starting from this page, using **Python** and the **BeautifulSoup** package, the complete list was scraped from a table using HTML attribute selectors and regex patterns. The municipalities from the Bolzano province that were present as well in the table have been left out. With this initial gathering, information such as name, population, density and altitude have been collected. Moreover for every entry of the table the link for the municipality own wikipedia page has been collected. From this each single page have been scraped in order to obtain additional data such as postal code, istat code, latitude, longitude and bordering municipalities. Only the bordering municipalities that are municipalities of Trento (that is only the ones included in the dataset) have been considered. For the initial cleaning processes the **Polars** library was used for manipulating the csv file with high performance and ease of use. Below we can see the structure of the end results.

166 rows

Column	**Description**
municipality	name of the municipality
province	name of the province
population	pop. according to ISTAT in the date 01/01/2024
surface	surface area in km^2
density	population density in ab/km^2
altitude	altitude expressed in meters from the sea level
wikipedia link	link of the municipality wikipedia page
postal code	a unique identifier related to a geographical area, used for sending mails or packages
istat code	unique identifier given by ISTAT
website	the official website od the municipality
bordering municipalities	array of bordering municipalities (*in practice an array of wikipedia links joined by a separator and represented as a string*)
latitude	latitude expressed like the following format: 46°09′N
longitude	longitude expressed like the following format: 11°12′E
latitude numeric	latitude expressed as a floating point number
longitude numeric	longitude expressed as a floating point number

2.5.2. Vivoscuola

Vivoscuola is the Trentino school website with the goal of supporting the ongoing improvement of the Trentino education system and the contact between schools and families. This website contains of all the schools from kindergarten to high schools (universities are not included) in the Trento province. A .csv file was easily available and downloadable right from the website. It required a little bit cleaning, that was done with **Python** and **Polars**, and it contained almost all of the data properties that will be listed below.

Now in order to access the "Sistema informativo della scuola trentina" APIs we needed an unique school id that could be found on the Vivoscuola school page, inside the HTML code. Using **Python** and **BeautifulSoup** we first scraped the search page, after artificially changing the query parameters in order to obtain the full list of institutes and schools. With this process we gathered all the school page links. Now we scraped each page in order to get the API id, in order to speed up the process we used a **multithreaded** approach.

1012 rows

Column	Description
main institute	name of the institute
school	name of the school (*empty if it is the main institute*)
institute type	text describing the purpose of the institute (*"istituzione scolastica e formativa" / "scuola dell'infanzia" / "circolo di coordinamento pedagogico"*)
administration type	type of administration (*"paritaria/equiparata", "a carattere statale/provinciale"*)
manager	name of the manager (*"dirigente scolastico" in italian*)
director	name of the director (*"direttore" in italian*)
educational coordinator	name of the educational coordinator
address	address indicating where the building is located
municipality	name of the municipality where the institute / school is located
telephone	telephone number of the school / institute
fax	fax number of the school / institute
institute email	contact email of the institute
management email	contact email of the management of the school / institute
secretary's office email	email of the secretary's office
website	official website of the school or institute
miur code	unique identifier given by the ministry of education (MIUR)
vivoscuola institute link	link of the institute (*or the institute from which the school belongs*) from the vivoscuola platform
vivoscuola school link	link of the school from the vivoscuola platform (*empty if it's an institute*)
entity	either *school* or *institute*
school type	a string indicating the kind of school (*"infanzia", "primaria", "secondaria di primo grado", "secondaria di secondo grado"*)

2.5.3. Sistema informativo della scuola trentina

Sistema informativo della scuola trentina is a digital service provider for schools in the Trentino region. It provides a digital registry for the professor as well as other tools for the managment in order to facilitate the supervision of the necessary operations. This system provides open API endpoints that can provide data regarding the number of classes and students attending each school.

After obtaining all the unique school ids from the vivoscuola website we are able to query the APIs exposed by the platform. Please note that the school link has been noted down in order to link this data with the vivoscuola dataset later. After some brief data cleaning the result is an array where every object is shaped like the following example (*given the fact that the data contains non-tabular nested objects it's not possible to create a csv file for now*):

```
{
        "school link": "https://www.vivoscuola.it/content/view/full/29266",
        "api id": "0222059697",
        "grades": {
            "percentage admissions total": 75.24557956777997,
            "results": [
                ...
                {
                    "percentages": [
                        61.76470588235294,
                        67.83216783216784,
                        91.80327868852459,
                        95.0
                    ],
                    "name": "Amessi 2023/24"
                },
                {
                    "percentages": [
                        64.8936170212766,
                        79.5774647887324,
                        79.81651376146789,
                        87.14285714285714
                    ],
                    "name": "Amessi 2024/25"
                },

            ]
        },
        "students": {
            "current_students": [
                {
                    "academic year": "2025/26",
                    "school year": 1,
                    "number of students": 173,
                    "number of classes": 8
                },
                {
                    "academic year": "2025/26",
                    "school year": 2,
                    "number of students": 159,
                    "number of classes": 9
                },
                {
                    "academic year": "2025/26",
```

```
                "school year": 3,
                "number of students": 134,
                "number of classes": 9
            },
            {
                "academic year": "2025/26",
                "school year": 4,
                "number of students": 62,
                "number of classes": 6
            }
        ],
        "past_year_students": [
            ...
        ],
        "past_2years_students": [
            ...
        ]
    }
}
```


2.6. Qualitative Evaluation

During this initial phase of purpose definition and information gathering the informal purpose, competency questions and scenarios required no modification. On the contrary we preferred to add additional cleaning phases on the data to make sure it fits perfectly with our desired purpose. Of course the raw data is still the source of truth so we must also make sure to not implement any forced modification on the reality of the data.

3. Language Definition

The language definition phase is important cause it ensures clarity and consistency by addressing ambiguity and diversity in natural and domain specific languages. It enables accurate data annotation, minimizes misunderstandings, and supports seamless integration across systems. This fundamental step is crucial for effective communication and interoperability in complex, multilingual, or multi domain projects.

The outcomes produced during this phase can be found in the Github repository.

3.1. Concept Identification

We began by identifying all the most relevant concepts and those who could have an ambiguous interpretation. Moreover given the multi lingual nature of the project both english and italian versions for the same concept must be provided.

For this reason we relied on formal descriptions from wiktionary and wikipedia whenever possible, by providing a clear multi lingual language definition for the most relevant terms in order to avoid any ambiguity. Their respective URLs have been chosen as unique identifiers for the concepts.

During this phase, we did not add any object properties.

3.2. Dataset Filtering

A major filter process has already been conducted in the earlier phases of the project, moreover there were no practical reasons for conducting an additional filtering activity during this phase.

3.3. Language Teleontology

In this step of the project, the focus was on constructing the language teleontology. Using **Protégé**, we formalized the language by defining the list of EType terms, object property terms, and data property terms as an OWL file. This included aligning our language to one or more reference language teleontologies, ensuring consistency and semantic clarity. The teleontology was then constructed to encode the domain language representation of our specific purpose, capturing object and data properties. This work can be considered a solid foundation from which additional refinements will be applied as we will see in the next sections.

The OWL file can be found in the github repository.

3.4. Qualitative Evaluation

During this phase we briefly considered the possibility of translating some terms regarding schools that are present in the data from Italian to English (example: *scuola dell'infanzia*). Upon further reflections, in particular by focusing on scenarios and personas, we considered that our hypothetical user base is Italian speaking. Moreover some nuances of these technical terms were difficult to be fully represented by an english translation. For these reasons, and given the appropriate formal definitions we gave during this phase, we decided to keep these terms in the Italian language.

4. Knowledge Definition

4.1. ER Modeling

Developing an ER Model can be a useful abstraction for gathering ideas and improving the quality of later formal modeling. We can determine the necessary **Etypes**, and their **object attributes** as well as **data properties** based on our Competency Questions and the acquired datasets. From Figure 3 we can clearly see the Etypes: Institute, School, Municipality, Grade Statistics, Student Information, their possible attributes and also the relations between them.

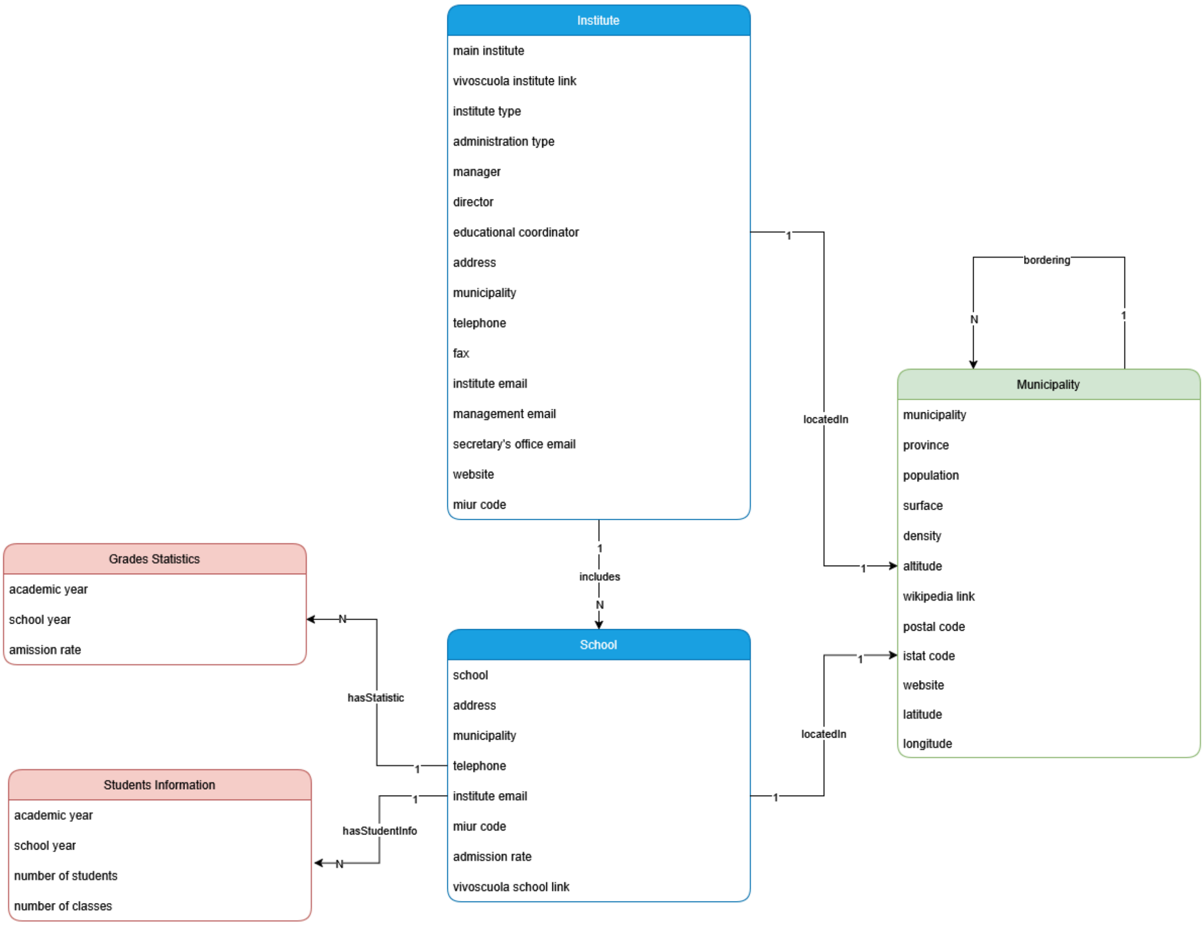

Figure 3: Representation of the ER model for the informal modeling phase

4.2. Knowledge Definition

The **Knowledge Definition** phase plays a central role in the iTelos methodology, as it focuses on formalizing and restructuring the gained information in order to begin building the final Knowledge Graph. During this phase, knowledge is organized into a coherent model through the design of teleontologies and the integration of multiple datasets. This chapter outlines the activities and outcomes achieved in this phase.

We relied on **Protégé**, a well established tool for ontology development, which supported the design and maintenance of the ontology by providing a comprehensive graphical user interface to define entity types, object properties, and data properties. We also drew inspiration from already existing concepts from *schema.org*, however because these entities came with already predefined properties it was not always possible to adapt each one of them according to our specific needs.

The results from this phase are reported below.

Class Name	**Description**
Thing > **Place**	Entities that have a somewhat fixed, physical extension. (*from schema.org*)
Thing > Place > **AdministrativeArea**	A geographical region, typically under the jurisdiction of a particular government. (*from schema.org*)

Class Name	Description
Thing > Place > AdministrativeArea > **Municipality**	The smallest civil administrative unit in Italy, also called "Comune" in Italian.
Thing > **Organization**	An organization such as a school, NGO, corporation, club, etc. (*from schema.org*)
Thing > Organization > **EducationalOrganization**	An educational organization. (*from schema.org*)
Thing > Organization > EducationalOrganization > **Institute**	Institutes (*"Istituti comprensivi"*) are administrative units that bring together one or more schools in order to ensure educational continuity within the same cycle of education.
Thing > Organization > EducationalOrganization > **School**	A single structure/school building in which teaching activities for a single school level (preschool, elementary, or middle school) take place.
Thing > **Intangible**	A utility class that serves as the umbrella for a number of 'intangible' things such as quantities, structured values, etc. (*from schema.org*)
Thing > Intangible > **StructuredValue**	Structured values are used when the value of a property has a more complex structure than simply being a textual value or a reference to another thing. (*from schema.org*)
Thing > Intangible > StructuredValue > **Grade_Statistics**	Information regarding the admission rates for a certain school, during a specific academic year and for a determined school year.
Thing > Intangible > StructuredValue > **Students_Information**	Information regarding the number of students for a certain school, during a specific academic year and for a determined school year.

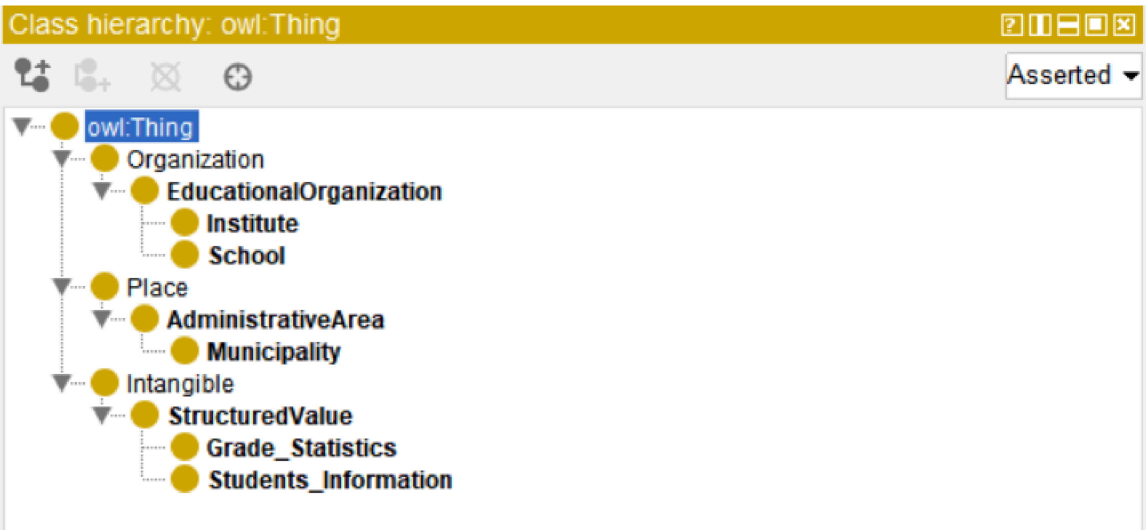

Figure 4: Examples of the schema definition from Protégé

Now let's go into the details for each class.

4.2.1. Municipality

Attribute	Description	Range
has_name	name of the municipality	*xsd:string*
has_province	name of the city where the municipality belongs	*xsd:string*
has_population	pop. according to ISTAT in the date 01/01/2024	*xsd:int*
has_surface	surface area in km^2	*xsd:float*
has_density	population density in ab/km^2	*xsd:float*
has_altitude	altitude expressed in meters from the sea level	*xsd:int*
has_postal_code	a unique identifier related to a geographical area, used for sending mails or packages	*xsd:string*
has_istat_code	unique identifier given by ISTAT	*xsd:string*
has_website	the official website od the municipality	*xsd:string*
has_latitude	latitude expressed as a floating point number	*xsd:float*
has_longitude	longitude expressed as a floating point number	*xsd:float*
has_link	link of the municipality wikipedia page	*xsd:string*
has_bordering_municipality	the municipality that shares the border with this one	*Municipality*

4.2.2. Institute

Attribute	Description	Range
has_name	name of the institute	*xsd:string*
has_institute_type	text describing the purpose of the institute (*“istituzione scolastica e formativa” / “scuola dell’infanzia” / “circolo di coordinamento pedagogico”*)	*xsd:string*
has_administration_type	type of administration, either private or public (*“paritaria/equiparata”, “a carattere statale/provinciale”*)	*xsd:string*
has_manager	name of the manager (*“dirigente scolastico” in italian*)	*xsd:string*
has_director	name of the director (*“direttore” in italian*)	*xsd:string*
has_educational_coordinator	name of the educational coordinator	*xsd:string*
has_address	address indicating where the building is located	*xsd:string*
has_telephone	telephone number of the institute	*xsd:string*
has_fax	fax number of the institute	*xsd:string*
has_email	contact email of the institute	*xsd:string*
has_management_email	contact email of the management of the institute	*xsd:string*

Attribute	Description	Range
has_secretarys_office_email	email of the secretary's office	*xsd:string*
has_link	link of the institute on the vivoscuola platform	*xsd:string*
has_website	official website of the institute	*xsd:string*
has_miur_code	unique identifier given by the ministry of education (MIUR)	*xsd:string*
has_school	a school that is part of the institute	*School*
has_location	the municipality where the institute is located	*Municipality*

4.2.3. School

Attribute	Description	Range
has_name	name of the school	*xsd:string*
has_address	address indicating where the building is located	*xsd:string*
has_telephone	telephone number of the school	*xsd:string*
has_email	contact email of the school	*xsd:string*
has_miur_code	unique identifier given by the ministry of education (MIUR)	*xsd:string*
has_link	link of the school on the vivoscuola platform	*xsd:string*
has_admission_rate	the average percentage of students that are admitted to the next class	*xsd:float*
has_school_type	a string indicating the kind of school *("infanzia", "primaria", "secondaria di primo grado", "secondaria di secondo grado")*	*xsd:string*
has_location	the municipality where the school is located	*Municipality*
has_statistic	statistics measuring the quality of the school during a certain point in time	*Grade Statistics*
has_students_information	information regarding the quantity of students in the school during a certain point in time	*Students Information*

4.2.4. Grade_Statistics

Attribute	Description	Range
has_academic_year	the academic year that this measure refers to	*xsd:string*
has_school_year	the school year (*e.g. the class*) that is measure refers to	*xsd:int*
has_admission_rate	the percentage of students that are admitted to the next class	*xsd:float*

4.2.5. Students_Information

Attribute	Description	Range
has_academic_year	the academic year that this measure refers to	*xsd:string*
has_school_year	the school year (*e.g. the class*) that is measure refers to	*xsd:int*
has_number_of_classes	the number of classes for this specific school year and academic year	*xsd:int*
has_number_of_students	the number of students for this specific school year and academic year	*xsd:int*

4.3. Qualitative Evaluation

During this phase, several refinements were introduced to improve consistency, integration, and compliance with the project guidelines. By starting from the ER model from the previous phases every entity name and property has been renamed in order to conform to the iTelos guidelines (for example by adding the *has_* prefix in the URI), but the overall structure has been maintained.

5. Entity Definition

The **Entity Definition** phase is the final step in the iTelos methodology, merging the knowledge and data layers into a unified Knowledge Graph. The input for this phase consists of cleaned and aligned data resources, as well as the teleontology created in previous stages. The objective is ensuring that entities across datasets are uniquely identified, matched, and mapped. The output of this phase includes the finalized KG and a set of mapping models in Turtle *.ttl* format.

5.1. Data Alignment

Data requires a final cleaning iteration in order to align the scraped content to the entity and knowledge definition presented above. For instance while parsing schools and municipalities both the Italian and German name were presented as one, or apostrophes and accents were often mixed, this made equality based equivalence quite difficult. Other minor tasks had to be completed such as renaming certain columns and values for clarity, and extracting object properties for later mapping. Using **polars** and **python** these errors where fixed and all the nomenclature was unified. Finally all the cleaned datasets where exported as *.csv* files.

This input data can be found in the entity definition folder on the github repository.

5.2. Entity Mapping

With the aid of **Ontotext Refine** we were able to efficiently implement the desired mapping with a very handy user interface and the GREL scripting language. In practical terms we were able to define custom Subject-Predicate-Objects statements starting from raw data, and modelling it according to our knowledge definition.Once the mapping has been obtained it was fairly easy to produce Turtle files and ingest them with the **GraphDB** engine for a later analysis.

All the Ontorefine mappings have been included in the relative Github folder.

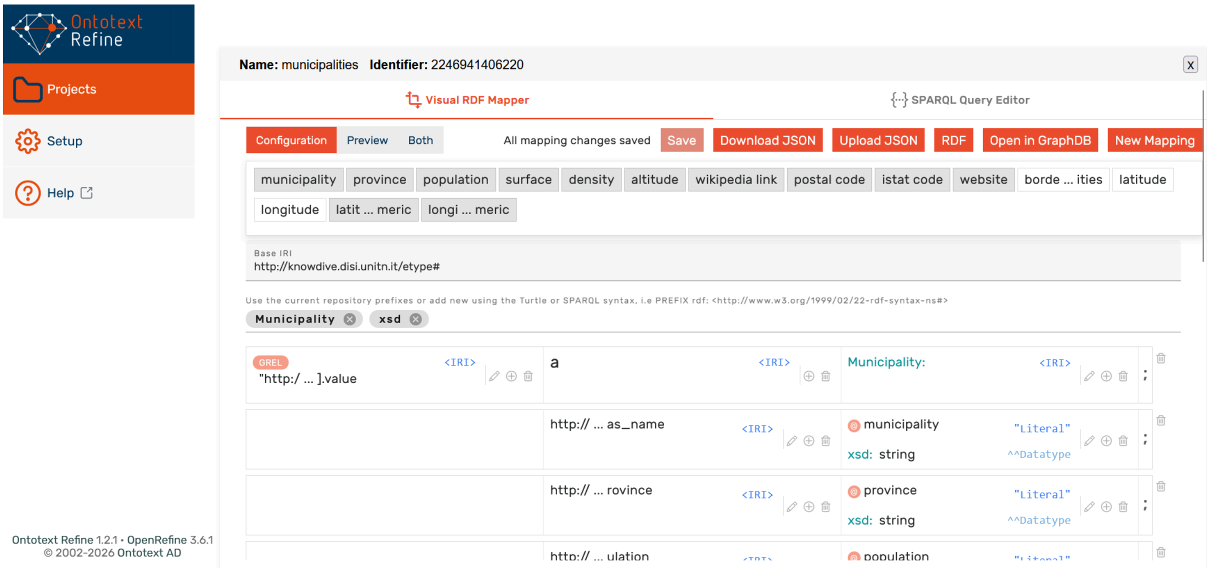

Figure 5: Examples of the mapping workflow from Ontotext Refine

5.3. Visualization

By using GraphDB's Explore functionality we tried to visually validate and inspect the structural correctness of the modeled entities and relationships, as well as to gain an overall understanding of the graph organization.

First we examined the relationship between classes, and we verified that the connections respected the intended ontology and semantic constraints.

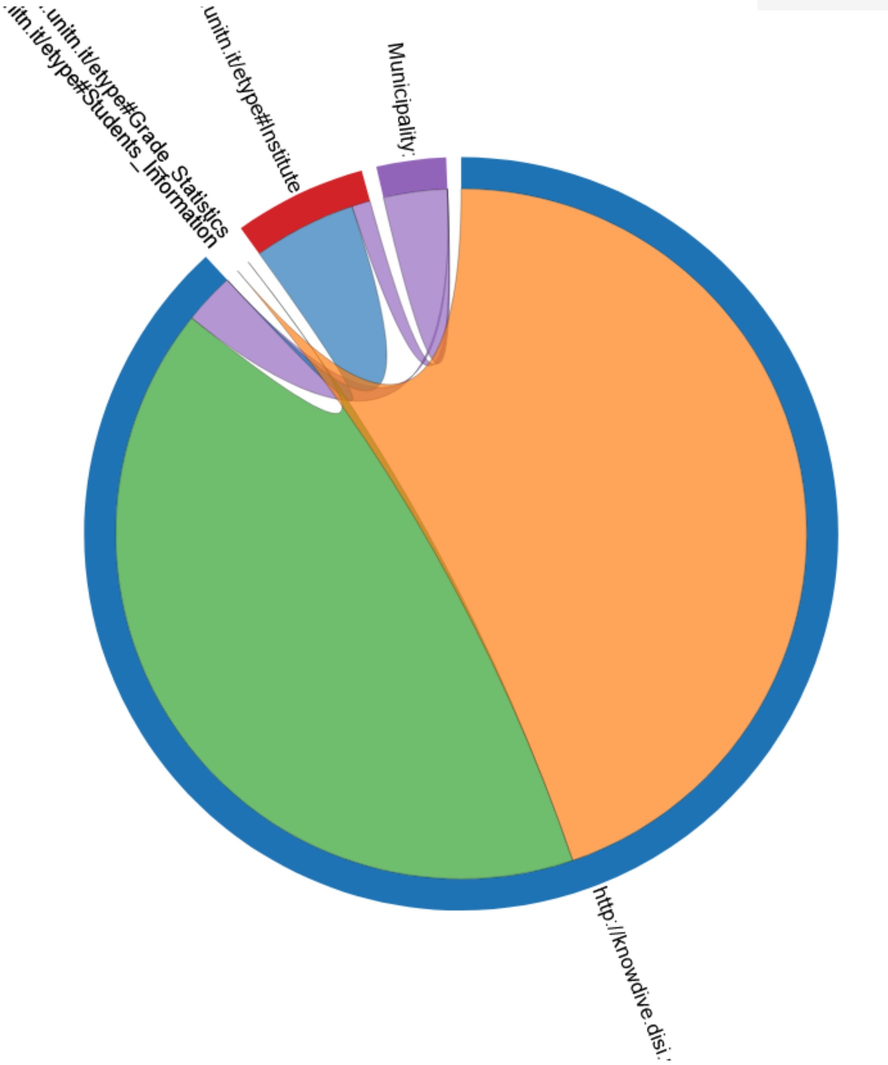

Figure 6: GraphDB representation of the relationships between different entities.

In addition, in the same section, GraphDB allows to visually explore certain subsections of the whole graphs. We went through a couple of examples for each entity in order to verify the structural correctness of the graph.

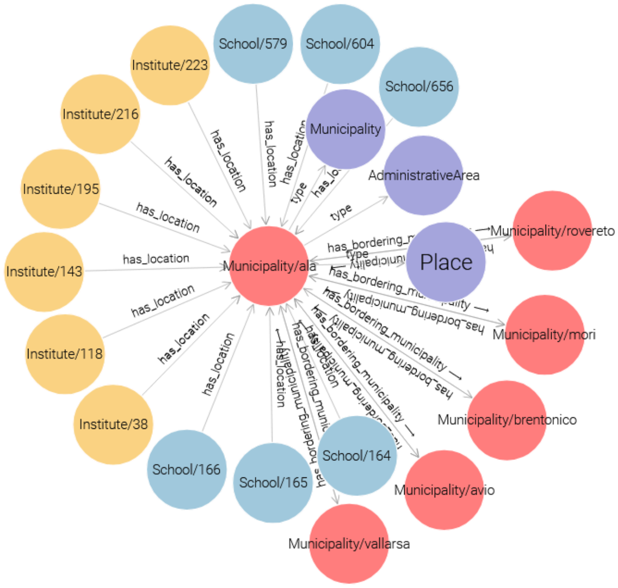

Figure 7: Visual representation of the municipality of “Ala” and its connected entities.

5.4. Qualitative Evaluation

During the entity definition phase, specifically while attempting to map the raw data using Ontorefine we noticed that most of the URI where incorrectly encoded due to a bug that occasionally can arise, in particular the “:” character was URL encoded as “%3A”. This was promptly fixed by deleting and recreating the column in the mapping definition.

6. Evaluation

6.1. Knowledge Graph Statistics

This paragraph outlines the main structural features of the completed Knowledge Graph, including the total number of entity types, properties, and their distribution.

In total we have 5 etypes: *Municipality, Institute, School, Grade_Statistics and Students_Information*

In terms of properties, the Knowledge Graph includes: **5 Object Properties**, this ensures that every class is connected one way or the other. This doesn’t imply that every class has exactly an object property in its domain but it means that either it has one in their domain or is the subject (range more precisely) of one. **48 Data Properties**, which capture the descriptive attributes of entities, such as names, properties, numerical measurements and many others. Together, these properties support both the structural connectivity of the graph and the detailed representation of individual entities.

Below is reported the number of entities for each etype
- 166 municipalities
- 269 institutes
- 743 schools
- 6330 grade_statistics
- 6913 students_information

6.2. Knowledge Layer Evaluation

A key aspect when evaluating the knowledge layer is the notion of coverage that is $Cov = \frac{\alpha \cap \beta}{\alpha}$ meaning a number between 0 and 1 measuring as the name says the level of coverage.

Regarding the **Etype Coverage** we have found a value of $Cov_E(CQ_E) = \frac{4}{5} = 0.8$ most notably because the property *admission_rate* of the *School* etype was found to be sufficient to answer those CQs which had the school quality in their concerns. Indeed an alleged user such as a public officer may be looking for historical data regarding the teaching quality so we wouldn't completely discard the *Grade_Statistics* class.

If we focus on the **Property Coverage** instead we find some highly usage of certain properties and some low usage for other properties. It is difficult to provide a precise numerical value because as a return statement we often preferred to return the whole Etype with all its properties by mimicking some hypothetical API retrieval for example. Nonetheless we can say that by looking at the were clauses of the SPARQL statements that almost every single object property and certain data property such as *has_name, has_school_type, has_altitude, has_longitude, has_latitude, has_population* are certainly being used extensively. Overall we believe the fact that some specific properties being overused in the filtering process is perfectly normal, not every property is suited for such task. There are indeed some properties that may be useful only for a very niche use cases but nonetheless we believe they shouldn't be discarded from the overall structure.

6.3. Data Layer Evaluation

6.3.1. Entity Connectivity

Entity connectivity measures the degree at which entities are connected to each other. Below we present the connectivity matrix, please note that due to the fact that *Municipality* can connect to itself, on the diagonal of the matrix are not counted non null data properties, but the number of connections just like the other cells.

	Municipality	**Institute**	**School**	**Grade Statistics**	**Students Information**
Municipality	920	0	0	0	0
Institute	269	0	743	0	0
School	0	0	0	6330	6913
Grade Statistics	0	0	0	0	0
Students Information	0	0	0	0	0

By considering $EC(X)$ Entity Connectivity for the EType X as $EC(x) = \frac{\sum_{Y=1}^{N}(X,Y)}{OP(x)}$ we provide the measure for each class:

- $EC(\text{Municipality}) = 920$
- $EC(\text{Institute}) = \frac{269+743}{2} = 506$
- $EC(\text{School}) = \frac{6330+6913}{2} = 6621.5$
- $EC(\text{Grade\_Statistics}) = 0$
- $EC(\text{Students\_Information}) = 0$

$EC(KG) = 8047.5$

6.3.2. Property Connectivity

By considering $PC(X)$ Property Connectivity for the EType X as $PC(X) = \frac{(X,X)}{DP(X)}$ we provide the measure for each class:

- $EC(\text{Municipality}) = \frac{166*13}{13} = 166$
- $EC(\text{Institute}) = \frac{3328}{17} = 195.76$
- $EC(\text{School}) = \frac{7342}{11} = 667.45$
- $EC(\text{Grade\_Statistics}) = \frac{6330*3}{3} = 6330$
- $EC(\text{Students\_Information}) = \frac{6913*4}{4} = 6913$

$EC(KG) = 14272.21$

We can notice that Institute and School may contain some null values in some of their data properties.

6.4. Query Execution

Finally we have successfully built our final knowledge graph, with a total of 166 municipalities and 1012 institutes and schools, including the relation between these entities and additional informations and statistics. Thanks to **GraphDB** we can fully explore it by using the **SPARQL** query language.

Below we've picked some examples from our queries, based on our **Competency Questions**, the full list can be found on GitHub.

6.4.1. Example 1

The following query (*cq n.3*) retrieves all the kindergartens from the municipality of "Lavis"

```
BASE <http://knowdive.disi.unitn.it/etype#>
PREFIX df: <http://knowdive.disi.unitn.it/etype#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

select ?s where {
    ?m rdf:type df:Municipality ;
    df:has_name "lavis" .
    ?s rdf:type df:School;
    df:has_location ?m .
    ?s df:has_school_type "infanzia"
}
```

	s	n
1	df:School/623	"scuola dell'infanzia di lavis "madre maddalena di canossa odv""
2	df:School/662	"scuola dell'infanzia di pressano"

Figure 8: Results from query n.3

6.4.2. Example 2

The following query (*cq n.6*) retrieves the top 20 schools located in the least populated municipalities

```
BASE <http://knowdive.disi.unitn.it/etype#>
PREFIX df: <http://knowdive.disi.unitn.it/etype#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

select ?s ?population where {
    ?m rdf:type df:Municipality ;
```

```
    df:has_population ?population .
    ?s rdf:type df:School;
    df:has_location ?m
} ORDER BY ASC(?population) LIMIT 20
```

	s	n	m	population
1	df:School/269	"scuola dell'infanzia di luserna"	Municipality:/luserna	"267"^^xsd:int
2	df:School/75	"scuola dell'infanzia di ruffrè"	Municipality:/ruffré-mendola	"420"^^xsd:int
3	df:School/237	"scuola primaria ronchi valsugana"	Municipality:/ronchi%20valsugana	"451"^^xsd:int
4	df:School/669	"scuola dell'infanzia di ronchi valsugana"	Municipality:/ronchi%20valsugana	"451"^^xsd:int
5	df:School/50	"scuola dell'infanzia di fierozzo "himblring""	Municipality:/fierozzo	"464"^^xsd:int
6	df:School/322	"scuola primaria s.felice fierozzo"	Municipality:/fierozzo	"464"^^xsd:int
7	df:School/78	"scuola dell'infanzia di bieno"	Municipality:/bieno	"465"^^xsd:int
8	df:School/47	"scuola dell'infanzia di casatta"	Municipality:/valfloriana	"473"^^xsd:int
9	df:School/217	"scuola primaria casatta valfloriana"	Municipality:/valfloriana	"473"^^xsd:int
10	df:School/358	"scuola primaria samone"	Municipality:/samone	"546"^^xsd:int
11	df:School/120	"scuola dell'infanzia di cavedago "l'albero azzurro""	Municipality:/cavedago	"576"^^xsd:int
12	df:School/46	"scuola dell'infanzia di capriana "maddalena lazzeri""	Municipality:/capriana	"595"^^xsd:int
13	df:School/215	"scuola primaria capriana"	Municipality:/capriana	"595"^^xsd:int
14	df:School/239	"scuola primaria telve di sopra"	Municipality:/telve%20di%20sopra	"620"^^xsd:int

Figure 9: Results from query n.6

6.4.3. Example 3

The following query (*cq n.11*) retrieves all the elementary schools located in a municipality within a 20km range from "Trento".

The equirectangular approximation has been used for calculating the distance between two points. Although this formula is fairly simple it may not be the best regarding accuracy.

```
BASE <http://knowdive.disi.unitn.it/etype#>
PREFIX df: <http://knowdive.disi.unitn.it/etype#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX ofn: <http://www.ontotext.com/sparql/functions/>

select distinct ?s ?distance_km_approx  where {
    ?m rdf:type df:Municipality;
    df:has_latitude ?lat0;
    df:has_longitude ?lon0;
    df:has_name "trento" .
    ?m1 rdf:type df:Municipality;
    df:has_latitude ?lat;
    df:has_longitude ?lon;
    BIND( (?lon - ?lon0) * ofn:cos( ((?lat + ?lat0)/2) * ofn:pi() / 180 ) AS ?dx ) .
    BIND(  ?lat - ?lat0  AS ?dy ) .
    BIND(ofn:sqrt( (?dx * ?dx) + (?dy * ?dy) ) * 111.32 AS ?distance_km_approx) .
    ?s rdf:type df:School;
    df:has_school_type "primaria";
    df:has_location ?m1 .
    FILTER( ?distance_km_approx <= 20 )
} ORDER BY ASC(?distance_km_approx)
```

	s	name	distance_km_approx
1	df:School/1	"associazione r. steiner - primaria"	"0.0"^^xsd:double
2	df:School/152	"scuola primaria arcivescovile trento"	"0.0"^^xsd:double
3	df:School/157	"scuola primaria sacra famiglia trento"	"0.0"^^xsd:double
4	df:School/169	"scuola primaria mattarello"	"0.0"^^xsd:double
5	df:School/170	"scuola primaria romagnano trento"	"0.0"^^xsd:double
6	df:School/376	"scuola primaria "g. a. tomasi" villazzano"	"0.0"^^xsd:double
7	df:School/377	"scuola primaria "u. moggioli" povo"	"0.0"^^xsd:double
8	df:School/379	"scuola primaria "e. bernardi" cognola"	"0.0"^^xsd:double
9	df:School/380	"scuola primaria "r. belenzani" s.vito"	"0.0"^^xsd:double
10	df:School/381	"scuola primaria "r. zandonai" martignano"	"0.0"^^xsd:double
11	df:School/382	"scuola primaria casa serena"	"0.0"^^xsd:double
12	df:School/384	"scuola primaria "a. nicolodi" trento"	"0.0"^^xsd:double
13	df:School/385	"scuola primaria "d. savio" trento"	"0.0"^^xsd:double
14	df:School/386	"scuola primaria "g. b. de gaspari" trento"	"0.0"^^xsd:double
15	df:School/387	"scuola primaria ravina"	"0.0"^^xsd:double
16	df:School/389	"scuola primaria clarina trento"	"0.0"^^xsd:double

Figure 10: Results from query n.11

7. Metadata

Metadata files describing licenses, authors, categories, tags, information resources, languages, dataset types, etc. have been included in order to provide the necessary contextual information about the project itself. This information has been organized by category in three different files: **project metadata**, **language metadata** and **knowledge metadata**.

All the files can be found in the GitHub repository.

8. Open Issues

With this project we successfully learned how to implement a knowledge graph from scratch by focusing on every single step, from scaping and data collection to implementing queries. Learning how to use popular tools like **Protégé**, **Ontotext Refine** and **GraphDB** was undoubtedly a positive experience. While the web scraping phase presented its own set of challenges, like dealing with inconsistent data and align the raw information to the desired ontologies, it has been a useful lesson.

Indeed there is definitely room from improvement, additional data sources can be brought in, such as data from universities, professors, courses, etc. Moreover the geolocation capabilities could be enhanced by attaching transportation data or precise latitude and longitude information from services like Google Maps or OpenStreetMaps.

Nonetheless we showed how from a limited number of data sources, a compact but valuable knowledge graph representation was possible.